Essay

Unraveling the science of Unidentified Anomalous Phenomena (UAP): Scalar fields, gravitomagnetics, superluminal craft and the quest for a unified field theory

Dr. Andrew D. Morgan – Department of Education, Government of Western Australia
March 2025

Based upon the work of Drs. Louis DeChiaro, Lee Hively, and Andrew Loebl

Abstract

Unidentified Anomalous Phenomena (UAP) challenge conventional physics with behaviours like instantaneous acceleration and apparent teleportation. By applying extended electrodynamics (EED) and extended gravitomagnetics (EGM), we can better grasp these phenomena as the manipulation of energy within a unified field. Maxwell's extended electrodynamics proposes a scalar field behind these phenomena, enabling instant displacement and frictionless motion. Similarly, extended gravitomagnetic theories suggest alternative gravitational interactions to elucidate these dynamics. Exploring these concepts offers insights into advanced propulsion systems and unique electromagnetism-gravity interactions within a unified field. When designing and constructing trans medium superluminal crafts for interstellar transits, leveraging extended electrodynamics and gravitomagnetics allows for creative thinking while maintaining a foundation in Classical Electrodynamics (CED).

"The proposition arises that a unifying field may underpin the enigma of UAP, which due to erroneous mathematical constructs has been potentially overlooked until now."

Figure 1. Becoming a space faring race requires rethinking our approach to science, incorporating scalar quantities into mathematical equations, re-evaluating the fundamentals of physics and unlocking technologies that enable super-luminal transits between star systems.

1. Introduction

For over 80 years, legacy UFO or UAP programs have pursued a technology that defies conventional explanation. In addition, publicly the significant pop-culture connection and vocabulary often misses the mark, with a crucial scientific component remaining elusive. This hindrance, including using terms such as "**anti-gravity**" impacts our ability to comprehend inexplicable observations and interactions with these phenomena using established physics and mathematics.

Historically and in contemporary times, UAP features and behaviours, from structured craft (**Figure 1**) to luminous orbs **(Figure 2**), are interpreted within existing knowledge boundaries. Sudden accelerations, defiance of inertia, instant positional shifts, and apparent engagement with human consciousness all contribute to a persistent sense of misunderstanding the science. If our knowledge of science is lacking it is perhaps because something has been missed or misrepresented mathematically, which has in turn led to an inability to explain UAP.

Figure 2. A type of orb (left) and rod (right) and their form variations manifesting and transiting the sampling area while filming in cardinal directions in the 850 nm infrared spectra.

Minor mathematical inaccuracies can have profound implications for unravelling the mechanisms of the observable universe, including UAP. Discovering and rectifying such errors could unlock coveted technologies, magnifying the societal impact of these mathematical nuances. Societal progress has the potential to advance our civilization as a conscientious and spiritually aware society capable of responsibly harnessing technological advancements and moving out amongst the stars.

In the present study Heaviside's gravitomagnetic equations are compared to Maxwells equations for classical electrodynamics. Mathematical expressions for gravity and magnetism are explained and suggestions made about the impact on physics when incorporating a scalar function that is analogous to Maxwell's extended equations. Such revelations are explained in terms of a unified field theory. Its implications for understanding UAP and developing alternative propulsion technologies for interstellar transits are also discussed.

2. Discussion

UAP in the form of orbs may be utilizing or harnessing some sort of invisible field, enabling them to present or manifest the physics-defying feats that are observed (**Figure 3**). Feats such as instantaneous acceleration, changing their form or morphology, and disappearing and reappearing in a different location almost instantaneously are observed. Such observations have been well documented in the field and analyzed extensively.

Observational evidence that some sort of invisible field is being occupied and harnessed includes the presence of bisected rings around orbs, their apparent ability to distend or elongate their shape and form, and rings or spindles that seem to wrap around the orbs and spin or rotate at extreme speed (**Figure 4 and 5**). Furthermore, recorded observations on variation in surrounding electric and magnetic field intensity suggest that some sort of energy coupling within this invisible field is occurring.

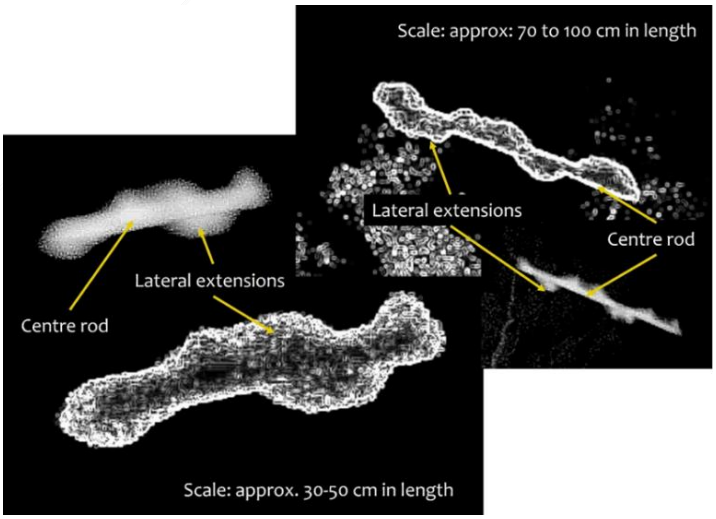


Figure 3. Image contrast and edge enhancement of extended rods showing a center structure surrounded by a rotating spindle(s) that can generate an energy field and hold an electric charge.

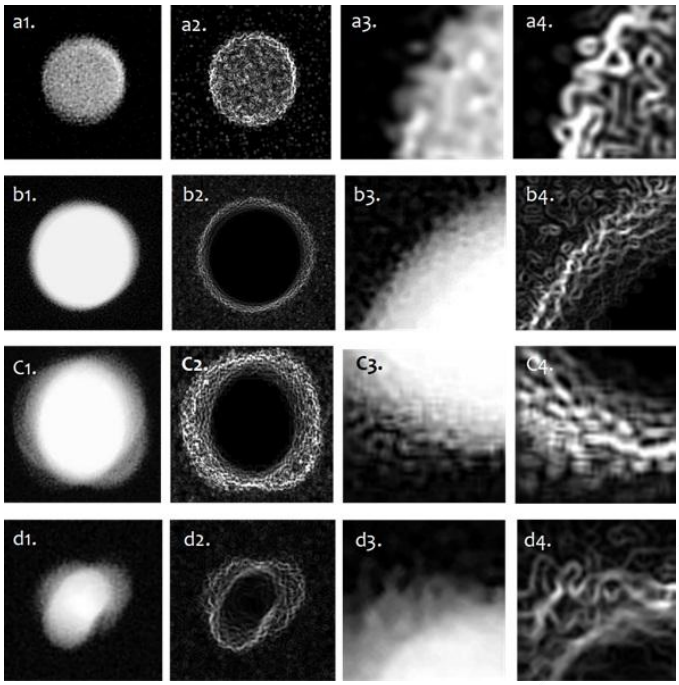


Figure 4. Image contrast, sharpen and edge analysis of the 4 types of orbs filmed in the 850 nm infrared spectra at 30 frames pers second: a2 to d2. Orb contrast, sharpen and edge enhancement, a3 to d3. Magnification of orb edges, a4 to d4. Contrast, sharpen and edge enhancement of magnification.

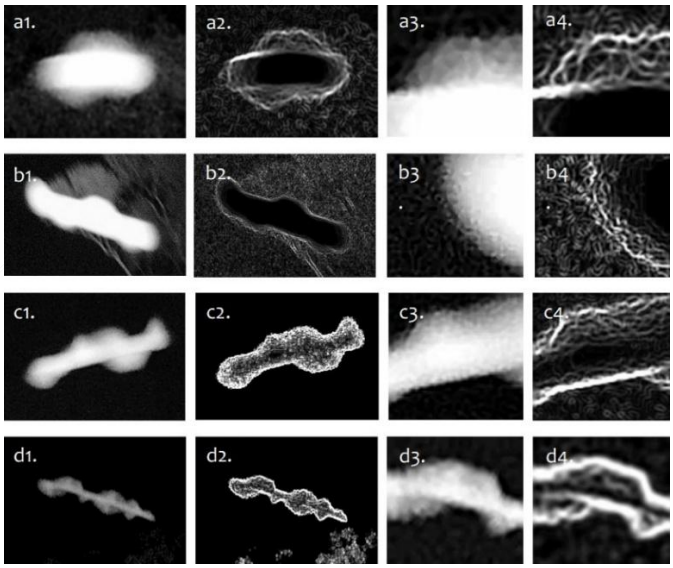


Figure 5. Image contrast, sharpen and edge analysis of the 4 types of rods filmed in the 850 nm infrared spectra at 30 frames pers second: a2 to d2. Rod contrast, sharpen and edge enhancement, a3 to d3. Magnification of orb edges, a4 to d4. Contrast, sharpen and edge enhancement of magnification.

The current understanding of electrodynamics, gravity and magnetism fails to explain adequately the manifestation of this phenomenon. Links between what is observed, the environment in which it is observed and what is known about physics, biology, chemistry and earth and space science fail to adequately explain what is going on.

However, in searching for answers it appears that an arbitrary decision long ago to ignore the scalar function in a key mathematical expression may be the root of our inability to adequately describe and explain these objects and indeed other UAP.

ρ = electric charge density

φ = scalar potential function

$\vec{J}$ = current density

$\vec{A}$ = vector potential

$\vec{\nabla}$ = vector differential operator

ε = dielectric properties

μ = magnetic properties

Currently, it is not possible to fully explain how UAP are able to do what they do using known science. However, it appears the answer to the dilemma may lie within erroneous mathematical constructs. In this regard, it is with James Clerk Maxwell and Oliver Heaviside that we must look for answers to the problem of UAP.

James Clerk Maxwell (1831 – 1879):

A Scottish physicist who was responsible for the first theory to describe electricity, magnetism and light as *different manifestations of the same phenomenon*.

Maxwell's equations for electro-magnetism (EM) have been called the "second great unification in physics," after the first one which had been realized by Sir Isaac Newton.

Oliver Heaviside (1850 –1925):

An English self-taught mathematician and physicist who rewrote Maxwell's equations in the form commonly used today. To simplify Maxwell's equations, Heaviside introduced a seemingly trivial error, but one which has had profound consequences.

The error in CED is the incorrect use of $\vec{\nabla} \cdot \vec{B} = 0$ by Heaviside, instead of Maxwell's original definition, $\vec{B} = \vec{\nabla} \times \vec{A}$, which is the solution to the former equation. **Heaviside's error eliminates the back-and-forth motion of "irrotational" or scalar fields**, while correctly retaining the closed loop "solenoidal" fields in CED.

2.1. CED and EED

It is an established fact that in classical electrodynamics the scalar component has been gauged away. In defining the electromagnetic spectrum as being only made up of the propagation of a transverse electromagnetic wave it has not been possible to define mathematically any other quantifiable field.

The error, a mathematical oversight by Heaviside, was not bought into question publicly. Afterall, gauging away a scalar function to zero simplified Maxwell's equations making them what they are today. Much has been gained in terms of their application in physics and the development of technologies that harness transverse electromagnetic wave propagation.

2.1.1. CED

The only solutions permitted in the differential CED equations shown below are the $\vec{E}$, $\vec{B}$, and propagation vectors that are all mutually perpendicular to each other (Hertzian waves) (**Figure 6**). All incidences where $\vec{E}$ *or* $\vec{B}$ are parallel to the propagation direction (longitudinal waves) are expressly forbidden.

Classical Electrodynamics (CED)

$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$	Faraday Law of Induction
$\vec{\nabla} \times \vec{B} = \mu\varepsilon \frac{\partial \vec{E}}{\partial t} + \mu\vec{J}$	Ampere-Maxwell Law
$\vec{\nabla} \cdot \vec{E} = \rho/\varepsilon$	Gauss E Law
$\vec{\nabla} \cdot \vec{B} = 0$	Gauss B Law
$\vec{B} = \vec{\nabla} \times \vec{A}$	Definition of Vector Potential, $\vec{A}$
$\vec{E} = -\overrightarrow{\nabla\varphi} - \frac{\partial \vec{A}}{\partial t}$	Definition of Scalar Potential, φ
$0 = \vec{\nabla} \cdot \vec{A} + \mu\varepsilon \frac{\partial \varphi}{\partial t}$	Lorenz Gauge

Figure 6. Maxwells' equations for classical electrodynamics, including the Faraday, Ampere-Maxwell and Gauss's laws, along with definitions of the vector and scalar potential functions.

However, the gauge condition "gauges away" the irrotational components of A, E, and J, all of which have been observed. Lightning is a simple example of an irrotational current (J), driven by an irrotational electric field (E), in turn driven by an irrotational vector potential (A), or magnetic field.

2.1.2. EED Mathematics

In redefining the scalar function in Maxwell's original equations, the irrotational components of A, E, and J, are reinstated (**Figure 7**). The mathematical expression of a field of 4 space geometric potential is expressed mathematically as a function of the scalar quantity. Crucially, the scalar function is no longer gauged to zero, but instead represents a field in and of itself, a scalar field.

Extended Electrodynamics (EED)

$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$	Faraday Law of Induction
$\vec{\nabla} \times \vec{B} - \overrightarrow{\nabla C} = \mu\varepsilon \frac{\partial \vec{E}}{\partial t} + \mu \vec{J}$	Ampere-Maxwell Law
$\vec{\nabla} \cdot \vec{E} + \frac{\partial C}{\partial t} = \rho/\varepsilon$	Gauss E Law
$\vec{\nabla} \cdot \vec{B} = 0$	Gauss B Law
$\vec{B} = \vec{\nabla} \times \vec{A}$	Definition of Vector Potential, $\vec{A}$
$\vec{E} = -\overrightarrow{\nabla\varphi} - \frac{\partial \vec{A}}{\partial t}$	Definition of Scalar Potential, φ
$C = \vec{\nabla} \cdot \vec{A} + \mu\varepsilon \frac{\partial \varphi}{\partial t}$	Scalar Field

Figure 7. Maxwells' equations for extended electrodynamics, including the new scalar functions for the Ampere Maxwell Law, Gauss's E law, and the scalar field 'C'.

EED corrects this error by including the irrotational components of the vector potential (A), electric field (E), and current density (J). Inclusion of the irrotational fields in EED requires two new terms in the dynamical equations of CED for Ampere's Law and Gauss' E Law. This then sets the Lorentz Gauge to equal C, rather than zero. We now have a complete mathematical expression of Maxwell's original equations, rather than applying an erroneous mathematical expression that negates part of the expression.

EED predicts at least three new types of waves:

- Scalar Waves (SWs): Purely scalar field-based waves.
- Scalar Longitudinal Waves (SLWs): Waves combining scalar fields with longitudinal electric fields.
- Helicoidal Waves (HWs): Waves involving scalar fields, magnetic fields, and longitudinal electric components.

2.2. CGM

In Heaviside's gravitomagnetic equations, like Maxwell's original formulation, the scalar function has been gauged away. The gravitational field arises from a scalar potential and a gravitational potential vector, resembling the configuration in electrodynamics, which consists of a scalar potential and electric field. Just like in electrodynamics, the gravitational gauge condition is enforced to eliminate the scalar potential.

Heaviside's development of the gravitomagnetic equations reflects his past oversight in applying vector calculus to classical electrodynamics, removing the scalar function. The "Lorenz gauge" condition in gravity is “mirrored”, ensuring the potentials adhere to a hertzian transverse wave equation format. As a result, the current limitations in physics emanate from a mathematical model that inaccurately depicts the observable universe as primarily defined by transverse electromagnetic wave propagation.

2.2.1. CGM Mathematics

Heaviside’s Classical Gravitomagnetics (CGM) equations are analogous to Maxwell’s CED equations and expand on Newton’s law of universal gravitation. These equations describe gravity using a gravitational field $\vec{g}$ (analogous to $\vec{E}$ in electromagnetism), a gravitomagnetic field $\vec{B}_g$ (analogous to $\vec{B}$), and the gravitational analogs of the vector potential $\vec{A}_g$ and scalar potential φ_g.

Gravitational Gauss’s Law
(Analogous to Gauss’s Law for $\vec{E}$)

$$\vec{\nabla} \cdot \vec{g} = -4\pi G \rho_g$$

The gravitational field $\vec{g}$ is generated by the presence of mass density ρ_g, like how an electric field is generated by charge density. The negative sign indicates that gravity is always attractive.

Gauss's Law for Gravitomagnetism
(Analogous to Gauss's Law for $\vec{B}$)

$$\vec{\nabla} \cdot \vec{B}_g = 0$$

There are no magnetic monopoles, and likewise, there are no gravitomagnetic monopoles (mass flow sources). The gravitomagnetic field $\vec{B}_g$ forms closed loops around mass currents.

Faraday's Law for Gravitomagnetism
(Analogous to Faraday's Law for $\vec{E}$)

$$\vec{\nabla} \times \vec{g} = -\frac{\partial \vec{B}_g}{\partial t}$$

A time-varying gravitomagnetic field $\vec{B}_g$ induces a gravitational field $\vec{g}$ with curl, just as a changing magnetic field induces a curl in the electric field.

Ampère-Maxwell's Law for Gravitomagnetism
(Analogous to Ampère's Law for $\vec{B}$)

$$\vec{\nabla} \times \vec{B}_g = -\frac{1}{c^2}\frac{\partial \vec{g}}{\partial t} - \frac{4\pi G}{c^2}\vec{J}_g$$

A changing gravitational field $\vec{g}$ induces a gravitomagnetic field $\vec{B}_g$. A mass current $\vec{J}_g$ (moving mass) also generates gravitomagnetic effects. This leads to frame-dragging effects observed in General Relativity (Lense-Thirring effect).

Definition of the Gravitational Vector Potential $\vec{A}_g$

$$\vec{B}_g = \vec{\nabla} \times \vec{A}_g$$

The gravitomagnetic field $\vec{B}_g$ can be expressed as the curl of the gravitational vector potential $\vec{A}_g$, just like in electromagnetism where $\vec{B} = \vec{\nabla} \times \vec{A}$.

Definition of the Gravitational Scalar Potential φ_g

$$\vec{g} = -\vec{\nabla}\varphi_g - \frac{\partial \vec{A}_g}{\partial t}$$

The gravitational field $\vec{g}$ comes from two sources: (1) the gradient of the gravitational scalar potential φ_g (like electrostatics); and (2) the time derivative of the gravitational vector potential $\vec{A}_g$ (analogous to the electric field in electrodynamics).

The Gravitational Gauge Condition
(Analogous to the Lorenz Gauge in Electromagnetism)

$$\vec{\nabla} \cdot \vec{A}_g + \frac{1}{c^2}\frac{\partial \varphi_g}{\partial t} = 0$$

This is the "Lorenz gauge" condition for gravity, ensuring that the potentials satisfy a hertzian transverse wave equation structure. Setting it to zero yields the same limitations as CED. The complete set of Heaviside's CGM equations:

Classical Gravitomagnetics (CGM)

$\vec{\nabla} \times \vec{g} = -\frac{\partial \vec{B}_g}{\partial t}$	Faraday Law of $\vec{B}_g$
$\vec{\nabla} \times \vec{B}_g = -\frac{1}{c^2}\frac{\partial \vec{g}}{\partial t} - \frac{4\pi G}{c^2}\vec{J}_g$	Ampere-Maxwell Law for $\vec{B}_g$
$\vec{\nabla} \cdot \vec{g} = -4\pi G \rho_g$	Gauss $\vec{g}$ Law
$\vec{\nabla} \cdot \vec{B}_g = 0$	Gauss $\vec{B}_g$ Law
$\vec{B}_g = \vec{\nabla} \times \vec{A}_g$	Definition of Vector Potential, $\vec{A}_g$
$\vec{g} = -\vec{\nabla}\varphi_g - \frac{\partial \vec{A}_g}{\partial t}$	Definition of Scalar Potential, φ_g
$0 = \vec{\nabla} \cdot \vec{A}_g + \frac{1}{c^2}\frac{\partial \varphi_g}{\partial t}$	Gravitational Gauge

Figure 8. Heaviside's equations for classical Gravitomagnetics, including the equivalent functions for the Faraday, Ampere-Maxwell and Gauss's laws, along with definitions of the vector and scalar potential functions.

2.2.2. EGM Mathematics

If the equivalent of the new scalar expression for C, and the new terms in the Ampere-Maxwell Law and Gauss's E law are applied in gravitomagnetics, gravity and magnetism are then linked within a universally present invisible field, a scalar field. We then have a scalar component for gravity and magnetism that is expressed mathematically, which can then be applied in physics. Einstein's general relativity can then be reconciled with Maxwells electromagnetism into one unified field theory.

If a scalar field C is applied to Heaviside's Classical Gravitomagnetics (CGM) equations, yielding "Extended Gravitomagnetics" (EGM) equations, even more types of scalar waves could be revealed. These new types of scalar waves may "couple" with Extended Electrodynamics (EED's) scalar waves, unifying electromagnetism and gravity with all other observable phenomena and forces; a unified field.

In terms of the physics of UAP, spin vectors representative of vortices and their generation through angular momentum could perhaps couple seamlessly with scalar wave propagation within this unified field. A link between the ability to occupy a 3-space vector field while simultaneously moving through a scalar field of 4-space geometric potentials could be established. An object or UAP would have the ability to manipulate gravity while also affecting spacetime.

Given this, questions proposed here about how UAP operate within this unified field consist of:

- Is gravity always "attractive" or could it be "repulsive", and
- Can there be "monopoles" as well as magnetic "dipoles".

In Classical Gravitomagnetics (CGM), the movement of mass generates gravitomagnetic effects, leading to gravitational thirring in General Relativity. The Lense-Thirring effect, also known as frame dragging, predicts how a slow rotating massive object affects spacetime, influencing nearby objects' orbits. Similarly, a fast-rotating small object could perhaps also affect spacetime if spun fast enough to achieve this. It is proposed here that reinstating the scalar function of EGM may allow for the mathematical expression of this.

Additionally, Extended Gravitomagnetics (EGM) introduces the possibility of magnetic monopoles, allowing for open-ended loops around a mass current. A magnetic monopole is a hypothetical point source of a magnetic field, possessing a single magnetic pole (either north or south), while a magnetic dipole is a system with two magnetic poles (north and south) separated

by a distance, like a bar magnet. While **electric monopoles** (single electric charges) exist, **magnetic monopoles** have not been observed. Magnetic fields are always described as dipoles.

Although it is suggested that UAP may make use of a magnetic dipole-double capacitor arrangement consisting of a stable inner layer and rotating outer layer to generate a repulsive energy field in physical space, the concept of a single monopole being able to do the same in a 4-dimensional space of scalar potentials warrants further investigation. In this case, in EGM a rapidly rotating small object could manipulate spacetime and simultaneously move in the opposite direction of its charge. An object could essentially pull itself along being attracted in the direction of its opposite charge within a repulsive energy field.

Within the scalar field the concept of anti-gravity loses relevance within a unified field theory, where a single mathematical expression explains seemingly inexplicable object behaviours, transcending the need for outdated terminology. By reincorporating the scalar potential function into EGM, the equations evolve, aligning with extended electrodynamics (EED) from Maxwell's equations. This integration unveils the connection between electricity, magnetism, and gravity, shedding light on how Unidentified Anomalous Phenomena (UAP) operate within a unified field.

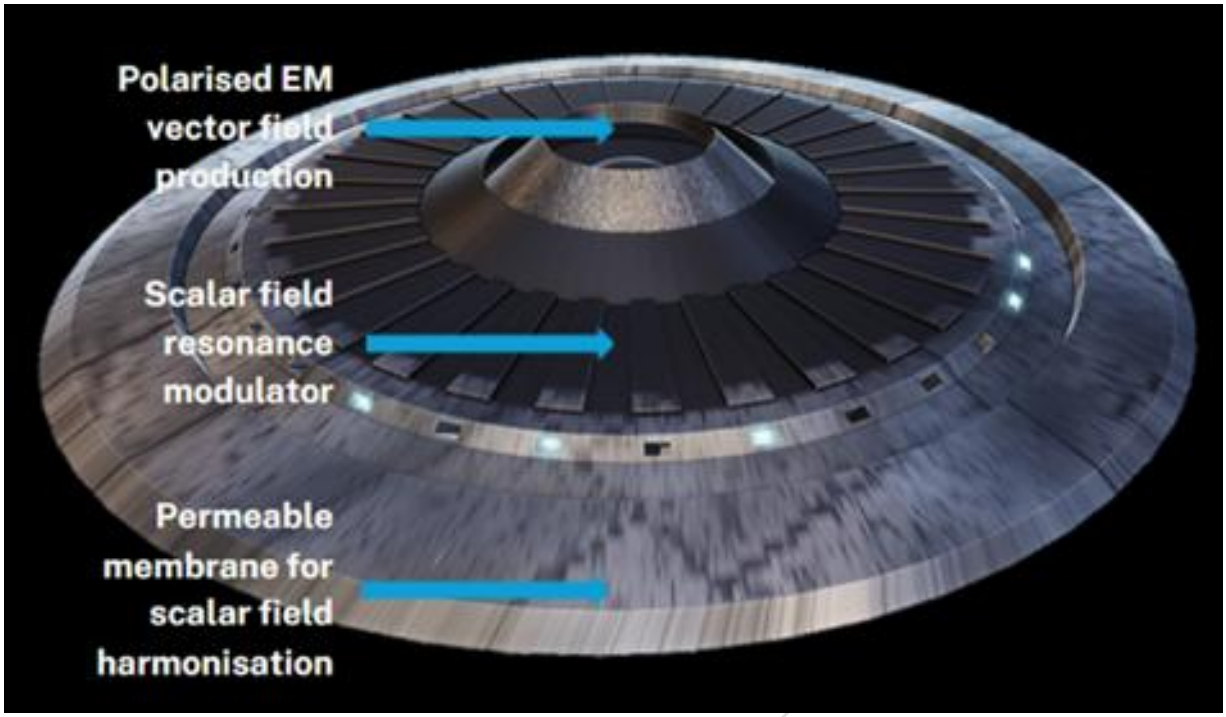
3. Conclusion

In applying extended electrodynamics (EED) and extended gravitomagnetics (EGM) to UAP, observations that defy known physics start to make sense (**Figure 9**). Firstly, in terms of the scalar function, for Maxwell's extended electrodynamics the following observations can be made:

- Disappearing then reappearing instantaneously at a different location implies harnessing some sort of scalar field, and
- Presenting with instantaneous acceleration implies being able to move effortlessly within this scalar field.

Secondly, in terms of the scalar field for Heaviside's extended gravitomganetics the following observations can be made:

- Presenting with no obvious airframe an objects structure must be such that it is able to produce the required energy to repulse gravity and overcome inertia, and
- The observed features of these objects suggest that some sort of magnetic dipole or monopole is produced to achieve this.



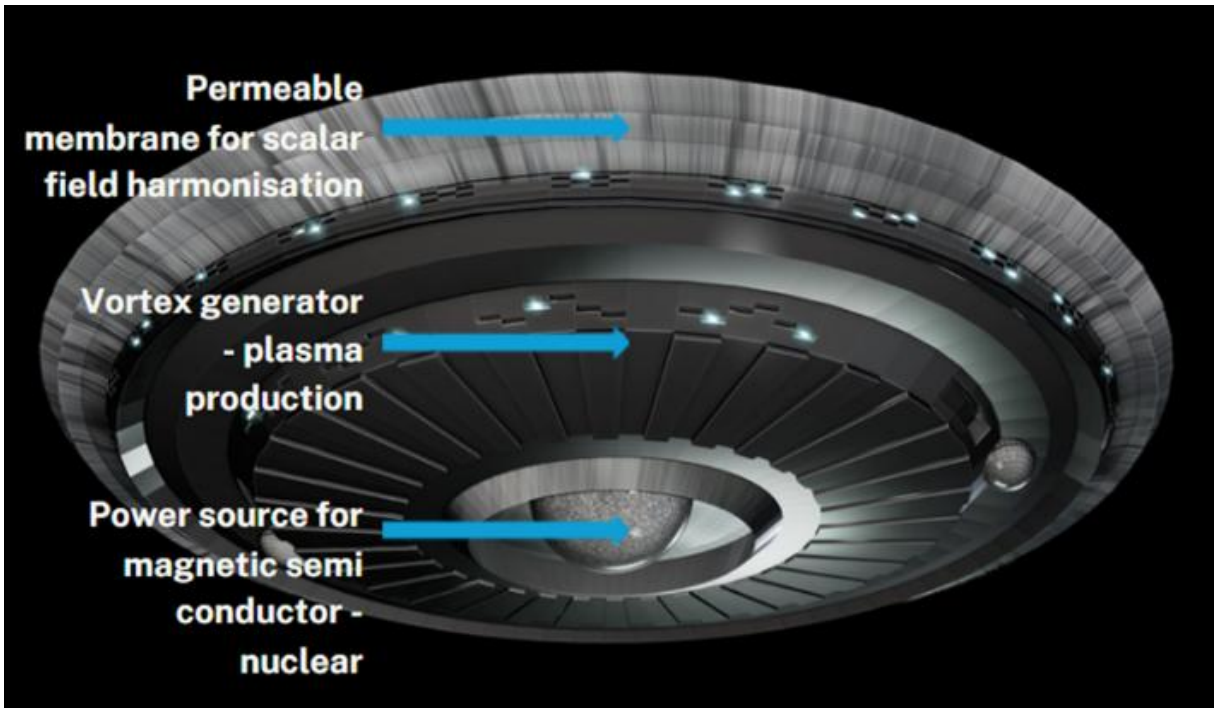


Figure 9: The ability of Unidentified Anomalous Phenomena (UAP) to seemingly defy known physics may be attributed to the properties of their outer layer, or 'skin,' which may interact with a scalar field.

For orbs and their associated forms, including rods, it appears that an inner structure is surrounded by an outer set of rings. By spinning at extreme speeds these outer rings may generate an energy field spin vector.

The differential between the inner and outer layer results in the creation of a magnetic dipole. Acting like a capacitor, an enormous buildup of energy is generated within the outer layer. Two things may be achieved, these being:

- Generation of a repulsive gravity field, and
- Generation of an enormous electric charge.

Such an arrangement does not necessarily rule out the monopole concept for UAP, particularly if it is crucial in being able to affect space time.

In conclusion, two emergent properties can be observed that are represented mathematically in Maxwell's and Heaviside's reinstated scalar functions for their equations, these are:

- Production of scalar waves that enable harmonization with a universally present scalar field, and
- The ability to manipulate gravity within this scalar field.

In both Maxwells and Heaviside's equations the coupling of the reinstated and mathematically scalar potential with gravity links Einstein's generation relativity with Maxwells electrodynamics, creating a unified field theory that may well provide an explanation for the observed physics defying observations of UAP.

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